MILITARY SPECIFICATION

CRUTCH, CANE TYPE, ADJUSTABLE, CHILD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an adjustable child's crutch, cane type, aluminum.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-225/6	Aluminum Alloy Bar, Rod, and Wire; Rolled, Drawn, or Cold Finished, 2024
QQ-A-225/8	Aluminum Alloy Bar, Rod, Wire and Special Shapes, Rolled, Drawn, or Cold Finished, 6061
QQ-A-250/8	Aluminum Alloy, 5052, Plate and Sheet
QQ-A-250/11	Aluminum Alloy, 6061, Plate and Sheet
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-R-566	Rod, Welding, Aluminum and Aluminum Alloys

FSC 6515

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SPECIFICATIONS

Federal

WW-T-700/6	Tube, Aluminum Alloy, Drawn, Seamless, 6061
ZZ-T-00351	Tip, Case and Crutch, Rubber
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed, and Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard

Military

MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-I-6866	Inspection, Penetrant Method of
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible
MIL-F-38762	Fluorescent Penetrant Inspection Units

STANDARDS

Federal

Fed. Test Method Std. No. 151	Metals: Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

DRAWING

Defense Personnel Support Center

24320	Crutch, Cane Type, Adjustable, Child

(Copies of specifications, standards, drawing, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY

Publication B3.0-41
Standard Qualification Procedure

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, NY 10017.)

3. REQUIREMENTS

3.1 Materials. Materials shall conform to the applicable portions of the specifications, standards, and drawing specified herein, and be subjected to the tests and meet the requirements specified therein. In addition, the metals shall conform to the chemical requirements of their applicable specifications when tested as specified in 4.5.1. When materials to be used are not specifically designated, those used shall be of the highest commercial quality and entirely suitable for the purpose intended.

3.1.1 Upper tube, lower tube, and handle. The upper and lower tube, and the handle shall be aluminum drawn seamless tubing conforming to WW-T-700/6, temper T6, Type I, composition 6061.

3.1.2 Pivot block and arm support post. The pivot block shall be aluminum bar conforming to QQ-A-225/6, Temper T4. The arm support post shall be 6061 aluminum rod conforming to QQ-A-225/8, Temper T6. As an alternate, the pivot block shall be 6061 aluminum conforming to QQ-A-225/8, Temper T6.

3.1.3 Arm support clip. The arm support clip shall be aluminum sheet conforming to QQ-A-250/8, 5052 - H32.

3.1.4 Crutch tip. The crutch tip shall be rubber conforming to size 18b of ZZ-T-00351. Color shall be black.

3.1.5 Hand grip. The hand grip shall be flexible vinyl conforming to the properties listed in Table I. Color shall be black.

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Table I. Properties of vinyl Hand Grip

Characteristic	Requirement
Specific gravity	1.30 to 1.40
Water absorption (24 hr), percent, maximum	0.5
Flammability	self-extinguishing
Tensile strength, psi, minimum	1,850
Elongation, percent (in 2 in.)	300 to 400
Hardness (Shore A)	70 to 80

3.1.6 Height adjustment and locking device. The device shall consist of two positioning buttons made from corrosion-resisting steel or steel treated to resist corrosion by cadmium or nickel plating, in accordance with QQ-P-416 or QQ-N-290 fitted to a carbon steel spring. As an alternate the buttons shall be attached to a flat spring steel strip (see drawing) bent into a "V" of required height.

3.1.7 Pivot pin. The pivot pin shall be stainless steel composition 301, 302, or 303. As an alternate, a stainless steel rivet shall be used, as shown on the drawing.

3.1.8 Welding rod. The welding rod used to weld the handle to the upper tube, and to weld the arm support clip to the pivot block shall be aluminum welding rod composition 5356 in accordance with QQ-R-566.

3.2 Construction. The crutch shall be single, telescoping aluminum tube construction with projecting plastic covered hand grip, and a C-shaped arm support. Bottom of crutch shall be fitted with a rubber crutch tip. Construction details shall conform to the drawing. Starting above hand grip, tubing shall bend slightly to provide comfortable positioning of arm support which shall be attached by means of an anchor block to an adjusting post telescoping into the upper end of tubing. Adjusting post shall be adjustable to three different heights. The lower portion of the crutch shall also be adjustable in length, to eight positions. The crutch shall conform to Defense Personnel Support Center Drawing 24320.

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3.2.1 Upper tube section. The upper tube shall be formed without wrinkling interior of the bend at upper end of tube. The upper end of the tube shall have three holes, drilled as shown to accommodate positioning pins.

3.2.2 Arm support post. The arm support post tube shall be an aluminum rod having a forked configuration to receive the pivot block of the support clip assembly. The tube shall provide a reasonably close fit within allowable tolerances when telescoped into the top of the upper tube. The tube shall have holes drilled to support the height adjustment locking device.

3.2.4 Arm support clip. The arm support clip shall be aluminum sheet dimensioned as shown on the drawing. One end shall be open to slip over forearm with opening edges rolled to the radius shown. The clip shall have a reinforcing web extending around the cuff as shown.

3.2.5 Pivot block. The aluminum pivot block shall be welded to the arm support clip. The block shall pivot freely on a corrosion resisting steel pin held in the arm support post by retaining pin clips (see drawing).

3.2.6 Handle and hand grip. The handle shall be an aluminum tube closed at one end and the other end welded securely to the upper tube below the bend and perpendicular to the tube. The handle shall be covered with a rubber or plastic hand grip secured at the end with a corrosion resistant screw and washer.

3.3 Finish.

3.3.1 Aluminum. Parts shall be anodized in accordance with type II of MIL-A-8625, to a uniformly dense oxide coating.

3.3.2 Corrosion-resisting steel. Parts shall receive no additional finishing beyond machine finish acceptable in screw machine production.

3.4 Welding requirements. Welding shall be done by qualified welders using appropriate equipment in a qualified process. Qualification shall be in accordance with American Welding Society Standard Procedure B3.0-41 or equal (see 4.5.3). Welds shall be capable of developing the strength of the parent metal. All welds shall be clean, nonporous, sound, and finished to match adjacent surfaces. All surfaces shall be free of sharp edges.

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3.4.1 Inspection of welds. There shall be no undercuts, overlaps, slag, flux, or weld spatter when subjected to visual inspection. Cracks or crack type defects shall not be permitted. If porosity is detected visually, penetrant inspection shall be utilized. Scattered surface porosity shall be the maximum acceptable, as revealed by the penetration method.

3.5 Identification marking. Each crutch shall be permanently and legibly marked with the manufacturer's name or registered trademark, placed in an inconspicuous location.

3.6 Workmanship. Workmanship shall be first class throughout. Crutches shall be well made, free of burrs and sharp edges, finished in appearance, and free from any defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

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4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, Inspection level II. Unit of product shall be one crutch.

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with referenced subsidiary specification. Level S-2 shall be used when no specification is referenced. Acceptance number shall be zero for all sample sizes.

4.2.3 For chemical analysis of the metals. Four ounces of each lot of each metal used in the construction of the items shall be selected for chemical analysis as specified in 4.5.1.

4.3 Examination. The crutches shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with, but not restricted to, the classification of defects listed in table I. Additional defects shall be classified in accordance with MIL-STD-105.

Table I. Categories and Defects*

Major	
101	Crutch not complete with all component parts.
102	Components not properly assembled or aligned.
103	Exposed and accessible edges and surfaces not free from sharp edges and burrs.
104	Dimensions not within tolerances specified.
105	Weld cracks, incomplete weld, excessive porosity, undercutting or overlaps.
106	Design and construction not as shown on the drawing.
Minor	
201	Aluminum parts not treated as specified.
202	Manufacturer's name or trade mark not located as specified, or omitted.
203	Weld slag, flux, spatter or dirt.

* AQL shall be 1.0 for major defects and 2.5 for minor defects.

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4.4 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table II. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table II. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and Packing Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape inproperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper location, sequence, or method of application.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements and subsidiary specifications. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.5.1 Chemical analysis of the metals. The samples selected in accordance with 4.2.1 shall be analyzed in conformance with methods 111 or 112 of Federal Test Method Standard No. 151. In case of dispute, method 111 shall be used for referee analysis. Non conformance shall be cause for rejection.

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4.5.2 Weld inspection. If visual inspection warrants, penetrant inspection of welds shall be conducted in accordance with either MIL-I-6866 or MIL-F-38762.

4.5.3 Qualification. In case of dispute of qualifications of welders or process, the Standard Qualification Procedure B3.0-41 and applicable portions of the Welding Handbook of the American Welding Society shall apply.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One pair of crutches, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each unit (one pair) shall be wrapped in kraft paper or other suitable material and packaged in a snug fitting box constructed in accordance with PPP-B-636, class domestic. Closure of the box shall be adequate to prevent accidental opening.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Twelve unit packages (12 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of the box specification.

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5.2.2 Level B. Twelve unit packages (12 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method I or II as specified in the appendix of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit packages. Each unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. The crutch is intended for use in ambulation by a child, temporarily or permanently disabled.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of packaging and packing.

6.3 Catalog information. The crutch covered by this specification is listed in the Federal Supply Catalog under the following stock number:

National Stock Number	Item Identification
6515-00-116-0563	Crutch, Cane Type, Adjustable, Child

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project Number 6515-1968
is

☆ U. S. GOVERNMENT PRINTING OFFICE: 1975-603-766/1203

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-37277(DSA-DM) CRUTCH, CANE TYPE, ADJUSTABLE, CHILD

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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